
Blocks, Elements and Modifiers (BEM)

— CSS Code Organization —

What is BEM

BEM (Block, Element, Modifier) is one of the most popular CSS naming conventions for classes

It's based on three essential components: blocks, elements, and modifiers

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Block

Block is a top-level abstraction of a new component

We can compare it to a container in which we put part of the code

The block should be a fully independent piece of the site

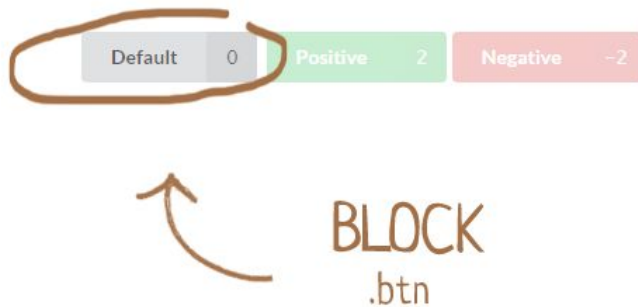
The main feature of the block is a possibility of re-using them anywhere on a page

Also, blocks can be nested inside any other blocks

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Block

Block is a top-level abstraction of a new component



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Element

A part of a block that has no standalone meaning and is semantically tied to its block

The element is a constituent part of a block that is assigned to it and can't be used outside of the block

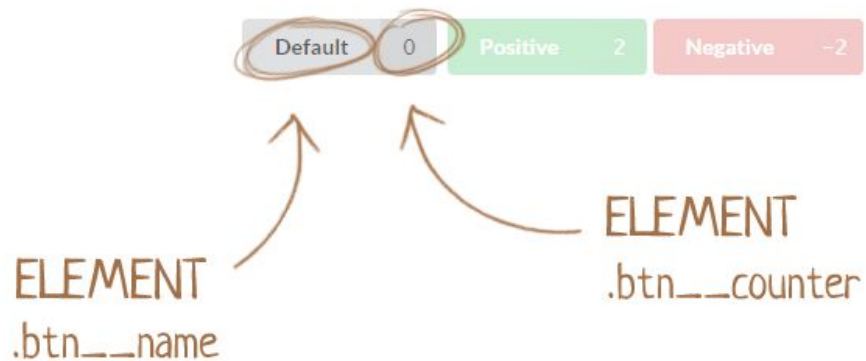
A name of the element consists block name and element name. The element name has to be separated by a double underscore (__)

Examples

`menu item, list item, checkbox caption, header title`

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Element



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Modifier

A flag on a block or element

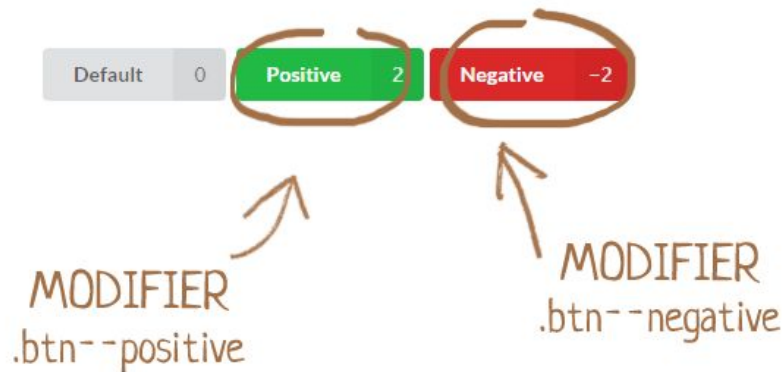
Use them to change appearance or behavior

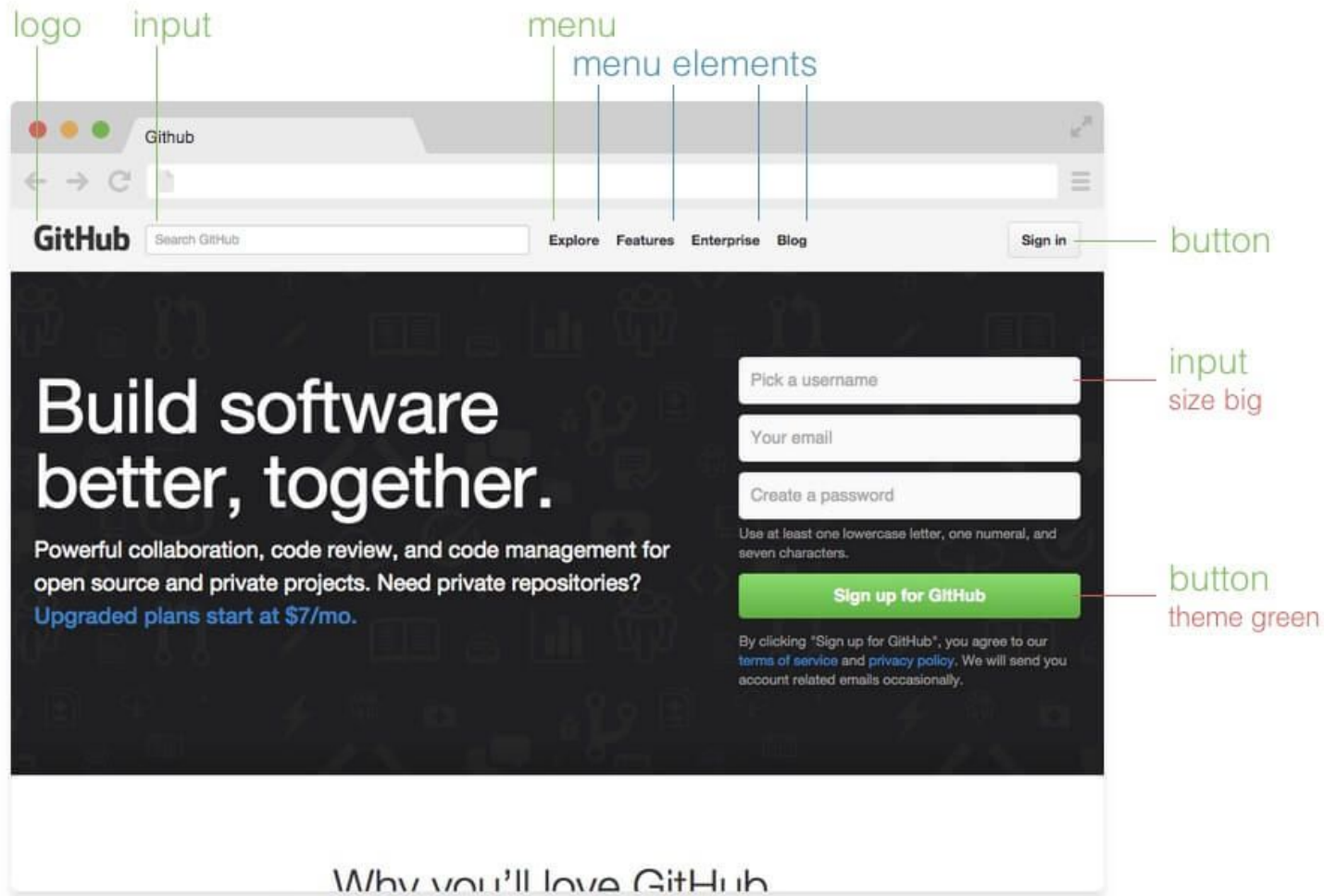
Examples

`disabled, highlighted, checked, fixed, size big, color
yellow`

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Modifier





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Let's look how one particular element on a page can be implemented in BEM



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```
<button class="button">Normal button</button>
```

```
<button class="button button--state-success">Success button</button>
```

```
<button class="button button--state-danger">Danger button</button>
```

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Normal button

Success button

Danger button

```
.button {  
  display: inline-block;  
  border-radius: 3px;  
  padding: 7px 12px;  
  border: 1px solid #D5D5D5;  
  background-image: linear-gradient(#EEE, #DDD);  
  font: 700 13px/18px Helvetica, arial;  
}
```

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```
.button--state-danger {  
  color: #900;  
}
```

Normal button

Success button

Danger button

```
.button--state-success {  
  color: #FFF;  
  background: #569E3D linear-gradient(#79D858, #569E3D) repeat-x;  
  border-color: #4A993E;  
}
```

Reference

<http://getbem.com/introduction/>

<http://www.smashingmagazine.com/2014/07/17/bem-methodology-for-small-projects/>

<http://csswizardry.com/2013/01/mindbemding-getting-your-head-round-bem-syntax/>

<https://www.merixstudio.com/blog/bem-code-organization-methodology#:~:text=BEM%20>

<https://en.bem.info/methodology/quick-start/>